



10CFR 50.73

CCN: 18-45

May 11, 2018

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Peach Bottom Atomic Power Station (PBAPS) Unit 2 and Unit 3
Renewed Facility Operating License No. DPR-44 and DPR-56
NRC Docket No. 50-278 and 50-278

Subject: Licensee Event Report (LER) 2-18-001

Enclosed is a Licensee Event Report concerning an unanalyzed condition regarding implementation of fire safe shutdown requirements. In accordance with NEI 99-04, the regulatory commitment contained in this correspondence is to restore compliance with the regulations. The specific methods that have been planned to restore and maintain compliance are discussed in the LER. If you have any questions or require additional information, please do not hesitate to contact Jim Kovalchick at 717-456-3351.

Sincerely,

A handwritten signature in black ink, appearing to read "Patrick D. Navin", written over a horizontal line.

Patrick D. Navin
Site Vice President
Peach Bottom Atomic Power Station

PDN/dnd/IR 4115309

Enclosure

cc: US NRC, Administrator, Region I
US NRC, Senior Resident Inspector
R. R. Janati, Commonwealth of Pennsylvania
S. Gray, State of Maryland
B. Watkins, PSE&G, Financial Controls and Co-Owner Affairs

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 60 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollect.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Peach Bottom Atomic Power Station Unit 2

2. Docket Number

05000277

3. Page

1 OF 4

4. Title

Fire Safe Shutdown Requirements Not Met for Spurious Operation of Valves at a High-Low Pressure Interface

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
03	14	2018	2018	- 001	- 0	05	11	2018	Peach Bottom Atomic Power Sta. Unit 3	05000278
									Facility Name	Docket Number
										05000

9. Operating Mode**11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)**

1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☒ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100%	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER**Licensee Contact**

James M. Kovalchick, Regulatory Assurance Manager

Telephone Number (Include Area Code)

717-456-3351

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
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14. Supplemental Report Expected**15. Expected Submission Date**☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No**Abstract** (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On 3/14/18, during a review of the Fire Protection Program Safe Shutdown Analyses, station personnel identified two valves that establish the boundary between a high pressure system and a low pressure system did not meet requirements contained in 10 CFR 50, Appendix R, Section III.G for protection against spurious operation during a fire. A flow path exists through the Reactor Water Cleanup (RWC) System into the Reactor Core Isolation Cooling (RCIC) pump discharge line into the 'B' feedwater line and into the low-pressure piping for the feedwater recirculation line back to the main condenser. During a fire, the cables could be damaged in such a way that the power cable to the outermost valve could be reenergized and the valve could open. The second valve is also normally closed but could spuriously open. If the fire also caused three other isolation valves in the RWC system to not close, a flow path to the main condenser could exist, resulting in a loss of reactor vessel water inventory.

This condition was not identified during the original analysis in the 1980's. There were no actual safety consequences as a result of this condition.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Peach Bottom Atomic Power Station Unit 2	05000277	2018	- 001	- 0

NARRATIVE**Unit Conditions Prior to the Event**

Unit 2 and Unit 3 were both operating in Mode 1 at approximately 100% rated thermal power. There were no structures, systems or components out of service that contributed to this event.

Description of Event

On 3/14/18, station personnel were reviewing and updating the Fire Protection Program Safe Shutdown Analyses. The update including classifying two motor operated valves in the feedwater system as "high/low pressure interface" valves. An analysis originally performed in the 1980's did not identify the valves as high/low pressure interface valves. It was determined that a flow path existed from the reactor vessel through the Reactor Water Cleanup (RWCU) System into the Reactor Core Isolation Cooling (RCIC) pump discharge line into the 'B' feedwater line and into the low-pressure piping for the feedwater recirculation line back to the main condenser. The two valves (in each unit) are:

Unit 2
MO-2-06-038B
MO-2-06-2663

Unit 3
MO-3-06-038B
MO-3-06-3663

FW Long Path Recirculation Isolation Valve
Long Path Recirculation Common Return Line

As high/low pressure interface valves, both valves must be assumed to have the potential to spuriously open due to fire damage to electrical cables for the valves.

A review performed in 2015 determined the two valves are a potential path for diversion of flow from the RCIC pump discharge if they should spuriously open during a fire, which could affect post-fire safe shutdown scenarios that rely on RCIC. Due to the potential for diversion of flow during a fire, power to the normally closed 2663/3663 valves was removed by opening the breaker during normal operations.

During the 2015 review, it was assumed that removing power from the closed valves during normal operations would prevent the valve from opening during a fire and would eliminate the risk of loss of reactor vessel inventory during a fire. It was determined that the only action needed was to update fire protection documentation and calculations to classify the valves as high/low interface valves.

During the March 2018 updating process, it was determined that the possibility of a 3-phase hot short during a fire also needed to be considered. The power cable between the breaker and the motor operator for the valve is in a cable tray with other energized cables. A 3-phase hot short could re-energize the valve operator and open the 2663/3663 valve. For this to create an actual flow path to the main condenser, fire damage to four additional valves would need to occur that would result in the valves spuriously opening or not closing if already open. The valves are the MO-2/3-06-038B valve and three normally open primary containment isolation valves in the RWCU system:

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Peach Bottom Atomic Power Station Unit 2	05000277	2018	- 001	- 0

NARRATIVE

MO-2/3-12-015	RWCU Inlet Inboard Isolation Valve
MO-2/3-12-018	RWCU Inlet Outboard Isolation Valve
MO-2/3-12-068	RWCU Outlet Isolation Valve

Note: the same valves exist in both units as indicated by "2/3" in the valve number.

Once this issue was identified, it was entered into the corrective action program. Because the valves had not been properly evaluated as high/low interface valves, compensatory measures were established in fire areas 6N for Unit 2 and 13N for Unit 3.

Further review determined the issue is also considered an unanalyzed condition and is reportable as required by 10 CFR 50.72(b)(3)(ii). See ENS event number 53265.

Analysis of Event

10 CFR 50, Appendix R, Section III.G requires at least one train of cables and equipment necessary for achieving and maintaining hot shutdown to be protected against fire damage. This includes damage from hot shorts, open circuits, or shorts to ground that could prevent operation or cause maloperation of such equipment. These requirements are further clarified in NRC guidance in Generic Letters 81-12 and 86-10. GL 86-10, Section 5.3.1 states:

For three-phase AC circuits, the probability of getting a hot short on all three phases in the proper sequence to cause spurious operation of a motor is considered sufficiently low as to not require evaluation except for any cases involving Hi/Lo pressure interfaces.

The exception for high/low pressure interfaces is due to the potential severity of a loss of reactor coolant event concurrent with a fire.

Section 6.2 of the site's FPP provides an analysis of high/low pressure interfaces that address the above Appendix R requirements. Fire Safe Shutdown Program Specification NE-00296, Section 6.1, provides program and design requirements for high/low pressure interfaces.

Cause of the Event

The flow path from the RWCU system to the feedwater return to condenser line was not identified as a high/low pressure interface during original Appendix R analysis in the 1980's. The exact cause is unknown, although it was likely due to lack of technical rigor in the system analysis performed when implementing Appendix R requirements.



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		2018	- 001	- 0

NARRATIVE

Corrective Actions

Since unprotected cables existed in fire areas 6N for Unit 2 and 13N for Unit 3, compensatory measures were conservatively established in accordance with Technical Requirements Manual (TRM) Section 3.14.8 Condition A. A design change has been developed to add a disconnect switch to the power cable for the 2663/3663 valve such that a 3-phase hot short could not occur. Additional corrective actions are documented in the corrective action program.

Previous Similar Occurrences

No previous similar occurrences have been identified.